

🏛️ Academic Employment

Aalto University

Postdoctoral Researcher, Advisor Dr. Samuel Kaski

June 2024–Present

Espoo, Finland

🎓 Education

Purdue University

PhD Statistics, Advisor: Dr. Anindya Bhadra

2019 – 2024

Indiana, USA

Thesis title: *On the use of α -stable random variables in Bayesian bridge regression, neural networks and kernel processes*

Universidad de Costa Rica

BS Pure Mathematics

2015 – 2019

San José, Costa Rica

Universidad de Costa Rica

BS Actuarial Science

2014 – 2017

San José, Costa Rica

📌 Research Interests

- Uncertainty Quantification
- Causal Inference

Applications:

- Bayesian inference
- Gaussian processes
- Personalized Medicine

- Computational methods
- Heavy-tailed distributions
- Environmental Sciences

📖 Publications

📄 Preprints

- » Fernández-Loría, C. and **Loría, J.** (2026+), Inferring Effect Ordering Without Causal Effect Estimation (*submitted*) [[arxiv](#)]

📄 Publications

- » Mäkinen L., **Loría, J.**, and Kaski, S. (2026), Bayesian Meta-Learning with Expert Feedback for Task-Shift Adaptation through Causal Embeddings (*in press, ICML 2026*) [[link](#)][[github](#)]
- » Björkman, Z., **Loría, J.**, Wharrie, S., and Kaski, S. (2026), Incorporating Expert Knowledge into Bayesian Causal Discovery of Mixtures of Directed Acyclic Graphs. *Proceedings of The 29th International Conference on Artificial Intelligence and Statistics*, in **PMLR** 300:964-972 [[link](#)][[github](#)]
- » **Loría, J.** and Bhadra, A. (2025), Deep Kernel Posterior Learning under Infinite Variance Prior Weights. *The 13th International Conference on Learning Representations (ICLR 2025)*. [[link](#)] [[github](#)]
- » **Loría, J.** and Bhadra A. (2024). Posterior Inference on Shallow Infinitely Wide Bayesian Neural Networks under Weights with Unbounded Variance. *Proceedings of the Fortieth Conference on Uncertainty in Artificial Intelligence*, **PMLR** **244** 2331-2349 [[link](#)] [[github](#)]
- » **Loría, J.** and Bhadra, A. (2024). SURE-tuned bridge regression. *Statistics and Computing* **34**, 30. [[doi link](#)] [[github](#)]
- » Viquez, J. J., Viquez, J., Campos, A., **Loría, J.**, Mendoza, L. A. (2018). Demographic Modeling via 3-dimensional Markov Chains *Revista de matemática Teoría y aplicaciones* available in <https://revistas.ucr.ac.cr/index.php/matematica/article/view/33608>

🏆 Awards

Somos Purdue Award

recognition of commitment to excellence and for leading with *ganas* at Purdue University.

Class of 2024

Bravo Award

one time staff award

Spring 2024

Ross-Lynn Research Scholar Grant

provides funding for notable graduate students

Spring 2024

Fully funded assistance to workshop on Bayesian Statistics and Statistical Learning

December 2023

IMSI, Illinois, USA

Department of Statistics Travel Award

Summer 2023

Department of Statistics Teaching Academy Associate Nominee

Spring 2023

Emily and Paul Kidwell graduate student excellence Award

Spring 2022

given to graduate students in the Department of Statistics in Purdue University interested in careers in academia.

Bob and Marjorie McLean Scholarship

Summers 2020 and 2021

given to distinguished graduate students of the Department of Statistics in Purdue University.

Bronze Medal

2015

awarded in Competencia Iberoamericana Interuniversitaria de Matemáticas

Honorable Mention

2014

awarded in Competencia Iberoamericana Interuniversitaria de Matemáticas

🗨 Talks and Posters Presented

Deep Kernel Posterior Learning under Infinite Variance Prior Weights

April, 2025

Poster in International Conference on Learning Representations

Singapore

Posterior Inference on Shallow Infinitely Wide Bayesian Neural Networks under Weights with Unbounded Variance

July, 2024

Poster in Uncertainty in Artificial Intelligence

Barcelona, Spain

Predicción en redes neuronales Bayesianas infinitas con pesos de varianza infinita

May 22nd, 2024

Seminario Intersede

CIMAT, Guanajuato, México

SURE-tuned Bridge Regression

August 7th, 2023

Contributed Paper in Joint Statistical Meeting

Toronto, Canada

SURE-tuned Bridge Regression

July 10th, 2023

Contributed talk in XVI Latin American Congress of Probability and Mathematical Statistics

São Paulo, Brazil

SURE-tuned Bridge Regression

May 26th, 2023

Poster in Statistics and Optimization in Data Science Workshop, in Purdue Business School

Indiana, USA

SURE-tuned Bridge Regression

April 2023

Invited talk in a weekly colloquium at CIMPA Universidad de Costa Rica.

Online

Learning the Ranking of Causal Effects with Confounded Data

October 2022

Flash talk in INFORMS

Indiana, USA

👥 Service to the Profession

PhD. Thesis advisor

» Lotta Mäkinen (ongoing, with S. Kaski as supervisor)

MSc. Thesis advisor

» Zachris Björkman (2024), Lotta Mäkinen (2025), and Axel Hedman (2025)

Reviewer

» Annals of Applied Statistics, AISTATS (2026), ICLR (2024, 2026), ICML (Silver Reviewer: 2026), Neurips (2023), TMLR (2026), Statistics in Medicine, UAI (2024, 2026)

🔧 Teaching Experience

Aalto University

Instructor

Fall 2022

Espoo, Finland

» Co-organized a seminar on causal inference, following Pearl's 'primer' book on causal inference.

Purdue University

Instructor

Fall 2022

Indiana, United States

» Taught one section of the course *Introduction to Statistics*. An introductory course for undergraduate students of computer science, mathematics, actuarial science, and statistics majors. Responsibilities included three weekly lectures, preparation of the relevant material, grading, and collaboration with the other instructors in the evaluation and design of the assignments of the course.

Purdue University

Instructor

Fall 2021 – Summer 2022, Summer 2023

Indiana, United States

» Taught two sections of the course *Elementary Statistical Methods*. An undergraduate level course for non-stem majors. Responsibilities included two weekly lectures to each section, preparation of the material, grading, leading teaching assistants and collaboration with other instructors for the evaluation of the course assignments.

Purdue University

Teaching Assistant

Fall 2019 – Spring 2021

Indiana, United States

» Held laboratories for two sections, office hours, and grading for the course *Elementary Statistical Methods*.

Universidad de Costa Rica

Instructor

Fall of 2017 & 2018

San José, Costa Rica

» Taught undergraduate course on basic functionalities of Excel and R. Responsibilities included lesson planning, course scheduling, and preparation of homework, exams, and course projects.

👛 Industry Experience

Banco Promérica

Financial Risk Analyst, analyzed financial data to evaluate and forecast credit risk

January 2019 – July 2019

San José Costa Rica

Quanto Soluciones S.A.

Actuarial Model Programmer, implemented credit risk models

January 2017 – July 2018

San José Costa Rica

💬 Communication

Spanish Native

English Advanced

Portuguese Basic

⚙️ Leadership Roles

- President of Statistics Graduate Student Organization

2021-2022

Indiana, USA

- Vice-president of Costa Rican Students Association at Purdue (CRISaP)

2021-2024

Indiana, USA